Coding Reports of Exploratory Research in Empirical Articles

Coding Manual

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This manual outlines details for coding research articles for the presence of reports of exploratory research. The primary project for this coding manual has been applied is, "Evaluating the Claim that Preregistration and Registered Reports Restrict Exploratory Research," https://osf.io/4vbtq/.

Importantly, this coding manual pertains to *reports* of exploratory research within articles, and does not directly address the presence of exploratory research at all. The latter is important, but very difficult–if not impossible–to detect without clear reporting.

Defining Exploratory Research

This manual is conceptually grounded in an early and prominent argument for preregistration, namely that it allows for a clearer separation between *confirmatory* analyses, or hypotheses generated prior to the study, and *exploratory* analyses, or analyses for which there were no prior hypotheses. Over the years, critics have raised concerns that implementing preregistration and requiring clear labeling of confirmatory and exploratory work would devalue and/or lead to a decrease in the prevalence of exploratory work.

For our purposes, we define exploratory research as, "any analysis that is indicated to be exploratory in nature or indicated to not be a planned test of a specific hypothesis," which is how the term is generally used in the debate about preregistration.

What Counts as Evidence for Exploratory Research?

The following will be coded as exploratory research: Any explicit mention of exploratory tests, exploratory analysis, unplanned analysis, speculative analyses. This includes tests labeled as robustness tests, sensitivity tests, or tests of alternative explanations that are explicitly or implied to be exploratory or unplanned.

Implied exploratory research can also be counted. This is a subjective criterion but is included here for cases where raters can determine that the analyses were exploratory absent an explicit statement. For example, if researchers state a research question, but indicate that they made no hypotheses about that question, then that would be considered exploratory even if they did not explicitly label it as such.

What Does Not Count as Evidence for Exploratory Research?

There are not any clear examples of seemingly exploratory research that would be categorically disqualified based on our definition. Not all uses of the word "explore" or "exploratory" necessarily indicate exploratory research. Rather, there are some situations where the context must be taken into account to judge whether or not the research was construed to be exploratory.

Two examples:

Following a hypothesized statistically significant interaction effect, authors may state that they "explored" or "probed" the interaction via simple slopes analysis. That kind of follow-up test is directly related to the hypothesis about the interaction and is conducted to gain greater clarity on the specific nature of the interaction. It also reflects the broader problem in psychology with the lack of specificity when proposing hypotheses about interaction effects (Baranger et al., 2023). It is generally not intended to be exploratory. The same logic goes for post-hoc tests in a hypothesized ANOVA or similar kind of analysis.

The use of "exploratory factor analysis" as an analytic technique is sometimes used in an exploratory way that is consistent with our definition, and sometimes used to test confirmatory hypotheses (despite its name). In both of these cases, the type of analysis could reflect exploratory research by our definition, but it will depend on the specific use in that study. Thus, the context will be taken into account when coding rather than simply examining the articles for keywords..

Procedure for Raters

PDFs of all articles are available in a shared Zotero library. Raters should open the PDF and conduct a Ctrl-F search for the following:

"explor", "plann", "a priori", "no hypoth", "no specific hypoth", "speculat", "robustness", "sensitivity", "follow-up", "alternative exp"

If none of these show up, raters should read through each of the relevant sections to assess potential evidence of reports of exploratory research. Once a unit receives a positive code for the presence of exploratory research, that unit no longer needs to be scanned.

Units to Which the Coding is Applied

All coding for exploratory research is absent (0) or present (1). For articles that contain multiple studies, the multiple sections that serve the same function (e.g., Results for Study 1, Results for Study 2) will be treated as one, as all studies will be registered or not registered. The preceding definition and decision rules should be applied to each of the article elements as follows:

Title - the title of the article

Abstract- the full abstract of the article

Introduction - only the end of the Introduction where the purpose of the current study is described. This could be labeled, "The current study," "The present study," "Research questions," "Hypotheses," or similar. Or it could not have a subsection header.

Method - the full section is to be examined, and special attention should be paid to a Data Analysis Plan subsection, if included. If a Data Analysis Plan is its own section, code it as part of the Method.

Results - the full section is to be examined

Discussion - the full section is to be examined

Section Headers Indicating Exploratory Research - Code absence (0) or presence (1) for the inclusion of section headers that explicitly indicate exploratory analyses. The full articles will be examined, but these are most likely to appear in the Results section when present.

Coding Difficulty - Provide a global assessment for how difficult it was to determine the ratings of each article. This serves as a useful indicator of the raters confidence in their ratings. We will use the system from O'Mahony (2023): 1 - Very easy, 2 = Somewhat easy, 3 = Somewhat difficult, and 4 = Very difficult. This is a global rating for difficulty, not for each section of the article.